

🍏 ADVANCED PYTHON · ESADE · FINAL PROJECT

iPhone Deal-Finder

Machine learning that predicts an iPhone's fair value and spots underpriced listings, on live data, in seconds.

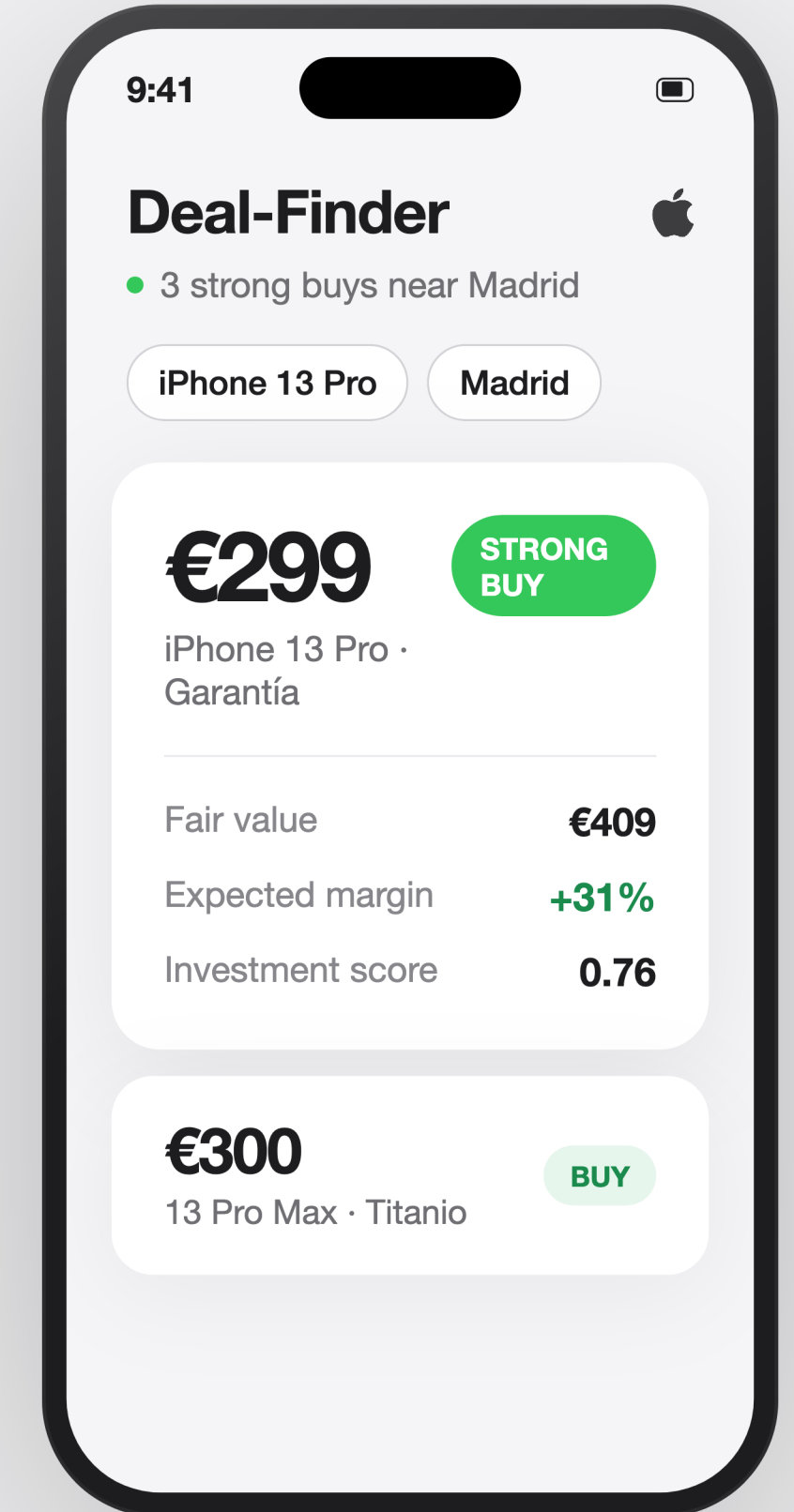
Regression

Web scraping

XGBoost

13,745 real listings

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The goal: a price brain for second-hand iPhones

- Question** “Of all iPhones for sale right now, which are the best deals to buy and resell?”
- Model** Supervised regression that predicts fair market value, plus calibrated uncertainty.
- On top** An investment score → a clear BUY / HOLD / SKIP decision, with reasons.
- Context** Wallapop is Spain's largest second-hand marketplace; prices are noisy and inconsistent.
- Real use** Resale arbitrage and smart buying · rank live listings by expected profit, flag scams.

Why it matters

€120 profit on a €420 phone

at a 28% margin · if you can tell a deal from a dud before it sells.

Deals vanish in hours

Speed and objectivity beat manual eyeballing.

THE PRODUCT

One command → ranked deals

```
$ python -m src.cli scan --model "iPhone 13 Pro" --city madrid --max-price 700
```

€299

iPhone 13 Pro · Tienda / Garantía

STRONG BUY

Fair value

Margin · confidence

Investment score

€409 [300–403]

+31.0% · MEDIUM

0.76

€300

iPhone 13 Pro Max 256 GB · Titanio

BUY

Fair value

Margin · confidence

Investment score

€569 [347–596]

+82.2% · LOW

0.73

Why this is genuinely hard

No labels

Wallapop shows asking prices, never sold prices · so the model learns fair value from the crowd.

Messy free text

Specs hide in Spanish prose: “batería 92%”, “como nuevo”, storage often omitted entirely.

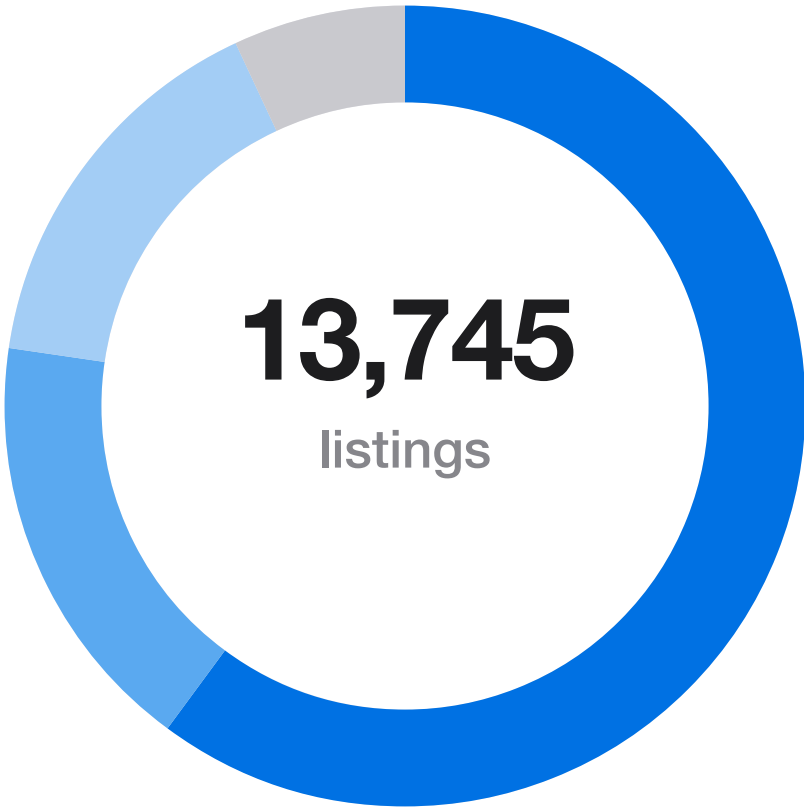
Bot walls

eBay and Backmarket fingerprint the TLS handshake and return 403 to datacenter IPs.

Uncertainty

A single price isn't enough; a recommendation needs calibrated confidence to be trusted.

13,745 real listings across 4 marketplaces



Wallapop

8,262

Scraped live from its JSON backend · no API, no keys. 20 models × 6 Spanish cities. The deployment market.



eBay sold

957

Real completed-sale prices with dates, via sold-listings HTML.



US e-commerce

2,361

2026 resale records, sale-evidenced.



eBay archive

2,165

Historical 2023 US listings for temporal depth.

Normalized · all prices → EUR, stored in Parquet, quality-weighted at merge.

System architecture



Single FeaturePipeline

Fitted once, serialized, reloaded at inference → identical features at train & serve. Zero training–serving skew.

Clean OOP

Abstract BaseScraper / BaseModel, `__slots__` containers, custom exception hierarchy, typed config, structured logging.

From Spanish prose to 105 features

Raw listing

iPhone 13 Pro 256GB Sierra Blue batería 92% como nuevo

↓ **parsed**

model · year	iPhone 13 Pro · 2021
storage	256 GB
battery	92%
colour	Sierra Blue
condition	like-new · 4/5

Regex + fuzzy

Model, storage, battery, colour, condition from free text · 97% model-ID rate.

TF-IDF

Top-50 description tokens capture wording signal.

Engineered

Days-since-posted, seller activity, photo count, city tier.

Market / source

Indicator features let one model serve multiple marketplaces.

Imputation

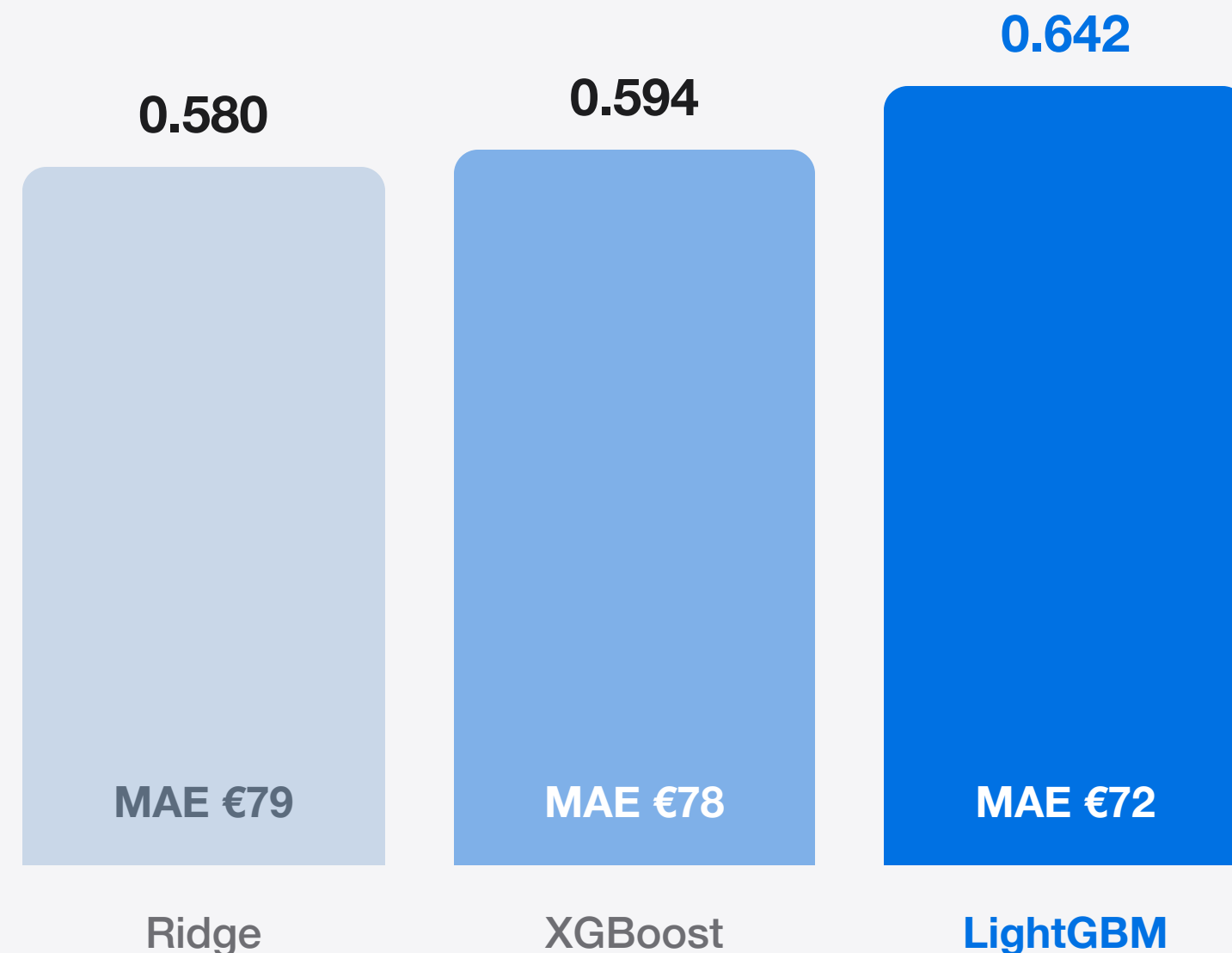
Battery / storage by model-family median; nothing dropped silently.

Baseline → boosting → calibrated intervals

Out-of-time split · no future leakage 80% intervals via Conformalized Quantile Regression

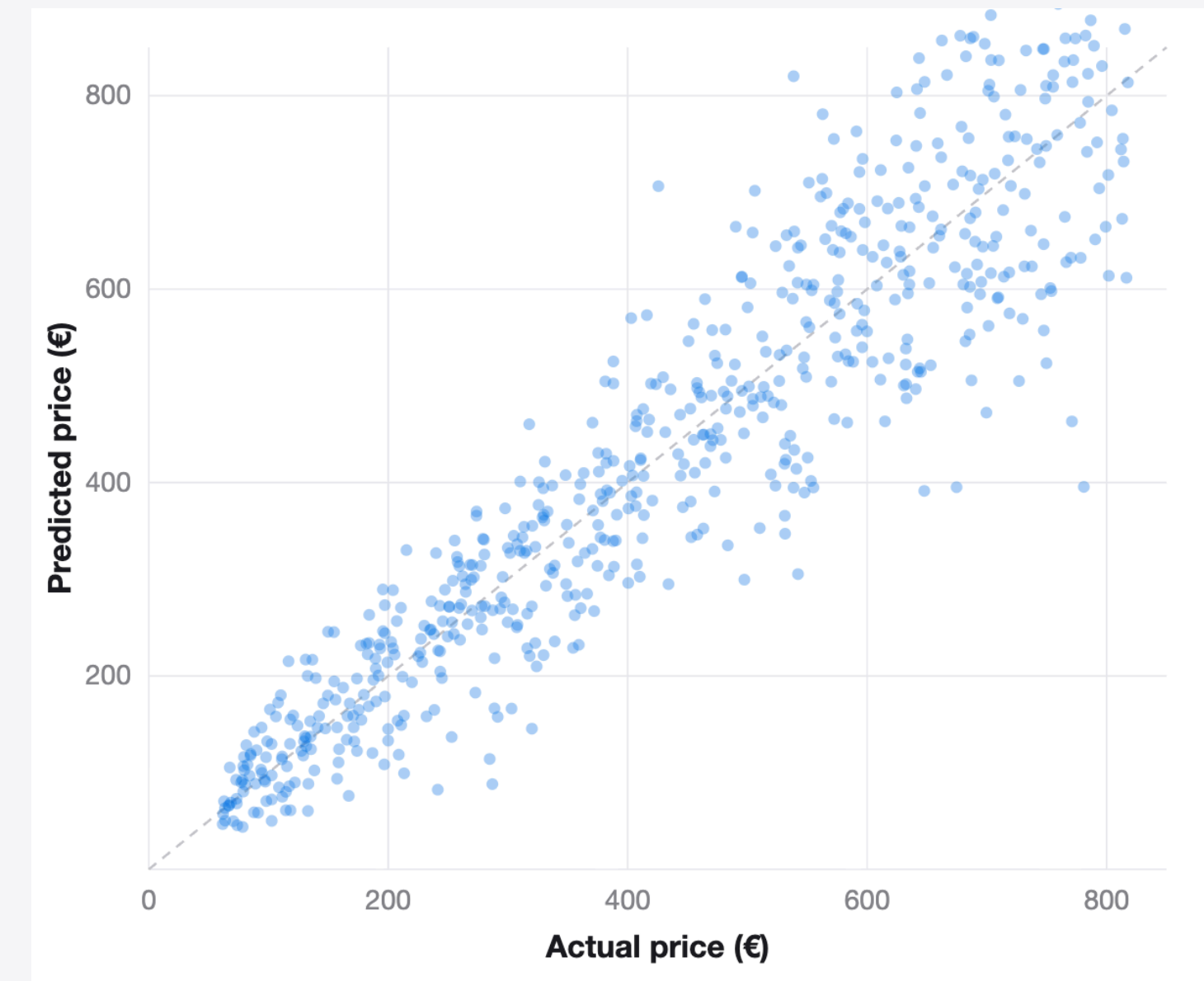
Held-out R^2 by model

MAE shown inside each bar



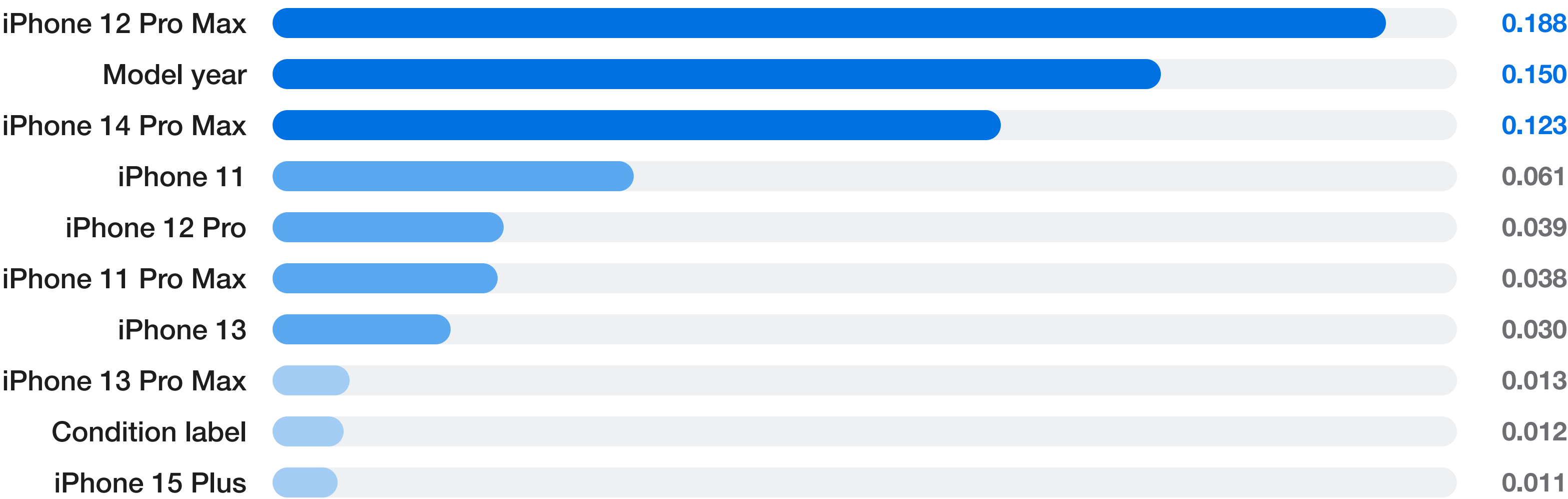
Predicted vs actual

LightGBM, held-out test set



What drives the predicted price

Top features by gain importance · model identity and release year carry the most weight.



Investment score → decision

```
investment_score =  
    0.5 · σ(margin · 5) + 0.3 · confidence +  
    0.2 · (1 - risk)
```

 **STRONG BUY** ≥ 0.75

 **BUY** $0.55 - 0.75$

 **HOLD** $0.35 - 0.55$

 **SKIP** < 0.35

Risk heuristics

- Price < 30% of fair value → **scam flag**
- No battery information
- Fewer than 2 photos
- “para piezas” / damaged wording
- iCloud-locked / suspicious

Why it beats eyeballing

Resellers

Rank hundreds of listings by expected profit in seconds · and never overpay.

Buyers

An objective fair-value second opinion plus a scam radar before you message a seller.

Researchers

A reproducible, versioned pipeline for second-hand price dynamics.

vs manual

Objective (model, not gut) · exhaustive (every listing) · fast (~1,670 listings/sec) · uncertainty-aware.

vs naive baseline

Ridge floor R^2 0.58 → boosting 0.64; calibrated intervals turn a guess into a confidence.

Six real problems, six fixes

Quantile intervals under-covered (60%)

→ **Conformalized Quantile Regression** → ~80%

eBay / Backmarket returned HTTP 403

→ **curl_cffi TLS (JA3) impersonation + cookie-warming**

eBay's sold search is capped

→ **diagnosed; scaled via narrow model×storage queries**

Merging US + ES data dropped R^2 0.74 → 0.61

→ **added market & source features; reweighted to ES**

SHAP crashed on XGBoost 2.x

→ **used XGBoost's native tree-SHAP (pred_contribs)**

Accessories / parts polluted the data

→ **title junk filter + damaged-device risk flag**

WHAT WE USED FROM THE COURSE

Concepts → code

S1 · OOP

Abstract classes · `__slots__`
· exceptions

S2 · Performance

Parquet · vectorized · `timeit`
(5.9× vs CSV)

S3 · Visualization

7 figure types across the
analysis

S4 · Parallel & Exceptions

Retry / backoff · custom
errors

S5 · Web Scraping

JSON API · `requests` HTML ·
TLS impersonation

S6 · Time Series

Out-of-time split ·
`TimeSeriesSplit`

S7 · Supervised ML

Ridge · XGBoost ·
LightGBM · CQR · SHAP

36 tests · 4 notebooks

Modular, packaged,
reproducible

Where this goes next

Counterfeit detection

A CNN on listing photos to flag probable fakes (S7 deep learning).

Time-to-sell

Predict liquidity · how fast a deal sells · as a 4th score dimension.

True panel data

Scheduled daily scrape to detect actual sales, price drops, time-on-market.

Productionize

FastAPI serve mode, CI on the test suite, GA hyperparameter search vs grid.

**Live data → fair value →
a confident buy/skip call.**

Academic project. The investment score is not financial advice · do your own due diligence.